Mutagenic and antioxidant activities of Croton lechleri sap in biological systems.

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The sap of Croton lechleri Muell.-Arg (Euphorbiaceae), called Dragon's blood, is used in folk medicine as a cicatrizant, anti-inflammatory and to treat cancer. In this research, the antioxidant activity of Croton lechleri sap was evaluated against the yeast Saccharomyces cerevisiae and against maize plantlets treated with the oxidative agents apomorphine and hydrogen peroxide. The mutagenic activity of the sap was also analyzed using the Salmonella/microsome assay (Salmonella typhimurium TA97a, TA98, TA100, TA102, TA1535) and in cells of the yeast Saccharomyces cerevisiae. The results showed that Croton lechleri sap possesses significant antioxidant activity against the oxidative damages induced by apomorphine in Saccharomyces cerevisiae under all the conditions studied. However, in the case of hydrogen peroxide, antioxidant activity of the sap was detected only in cells in the stationary phase of growth. The sap was also able to protect cells of the maize plantlets from the toxic effect of apomorphine. This sap showed mutagenic activity for strain TA1535 of Salmonella typhimurium in the presence of metabolic activation and a weak mutagenic activity for strain TA98. These strains detect base pair substitutions and frameshift mutations, respectively. Mutagenicity was also observed in a haploid Saccharomyces cerevisiae strain XV185-14c for the lys1-1, his1-7 locus-specific reversion and hom3-10 frameshift mutations.